Farzaneh Barat

Ph.D. Student in Mechanical Engineering

Lithium-Ion Batteries | **Battery Modeling** | **State Estimation** | **Machine Learning** | **Controls**

Lawrence, KS barat@ku.edu 785-424-2521

farzanehbarat.github.io Google Scholar LinkedIn GitHub

Technical Skills

Battery Modeling & BMS: Lithium-ion battery modeling, equivalent-circuit models, electro-thermal modeling, state-of-charge (SOC) estimation, parameter identification, degradation analysis, second-life battery systems, battery diagnostics, smart charging, thermal management

State Estimation & Controls: Nonlinear observers, Kalman filtering, Ensemble Kalman methods, particle filtering, state-space modeling, system identification, optimal control, numerical optimization

Programming & Machine Learning: Python, PyTorch, NumPy, SciPy, Pandas, MATLAB, Simulink, neural networks, neural ODEs, scientific computing, numerical integration, Monte Carlo simulation

Battery Testing & Experimentation: Battery cycling, charge/discharge testing, capacity characterization, HPPC testing, SOC characterization, thermal measurements, degradation testing, second-life battery-module evaluation, PEC battery cyclers

Engineering Software: ANSYS, Abaqus, SolidWorks, Autodesk Inventor, AutoCAD, CATIA, LaTeX

Education

University of Kansas **Lawrence, KS**
Ph.D. in Mechanical Engineering *Aug. 2023 – Present*

- **Advisors:** Prof. Huazhen Fang and Prof. Rachel Carter
- **Research Areas:** Lithium-ion battery modeling, estimation, parameter identification, electro-thermal systems, thermal management, machine learning, and second-life batteries

University of Kashan **Kashan, Iran**
M.Sc. in Mechanical Engineering *Sep. 2015 – Feb. 2018*

- Research in nonlinear optimal control and path tracking of robotic manipulators.

University of Kashan **Kashan, Iran**
B.Sc. in Mechanical Engineering *Sep. 2011 – Sep. 2015*

Battery, Controls & Machine-Learning Research

University of Kansas **Lawrence, KS**
Graduate Research Assistant *Aug. 2023 – Present*

- Develop physics-based and data-driven methods for lithium-ion battery state estimation, parameter identification, thermal management, degradation characterization, and second-life battery evaluation.
- Implement battery models, nonlinear estimators, machine-learning algorithms, numerical solvers, and experimental data-processing pipelines using **Python, PyTorch, MATLAB, and Simulink**.
- Design and analyze battery experiments involving dynamic charge/discharge profiles, capacity characterization, HPPC testing, thermal measurements, and degraded or second-life battery modules.

Neural-ODE-Enhanced Observer for Robust Battery SOC Estimation *2026 – Present*

- Developed a hybrid physics–machine-learning observer for lithium-ion battery SOC estimation under model-parameter uncertainty.
- Integrated a nonlinear electro-thermal battery observer with a **neural ordinary differential equation (neural ODE)** that learns the time evolution of SOC-estimation error.
- Implemented training and simulation in **PyTorch** using explicit Runge–Kutta integration and evaluated robustness across dynamic drive cycles, different C-rates, SOC levels, parameter mismatch, measurement noise, and inaccurate initial SOC.
- Designed a sparse-label training strategy requiring only a small number of SOC reference measurements along each trajectory, reducing the dependence on densely labeled experimental data.

Recursive Parameter Identification for Nonlinear Battery Models *2025 – 2026*

- Developed sequentialized Ensemble Kalman Inversion methods for recursive parameter identification in nonlinear stochastic state-space models with unmeasured internal states.
- Combined ensemble-based parameter updates with particle-based state inference for simultaneous treatment of parameter uncertainty and nonlinear state dynamics.
- Applied the framework to lithium-ion battery equivalent-circuit models using both simulated and experimental data.
- Evaluated parameter-estimation and state-estimation accuracy against nonlinear filtering and dual-estimation approaches.

Dynamic Thermal Management and Smart Charging of Lithium-Ion Batteries *2026 – Present*

- Investigate state-dependent thermal-management and charging strategies for lithium-ion batteries to reduce temperature nonuniformity, degradation, and lifetime loss.
- Study adaptive cooling strategies under varying SOC, charging conditions, external thermal disturbances, and dynamic operating environments.
- Support development of coupled battery electro-thermal models and experimentally informed control strategies for cylindrical and pouch cells.

Second-Life Lithium-Ion Battery Characterization and Modeling *2026 – Present*

- Characterize second-life lithium-ion battery modules through capacity testing, dynamic cycling, HPPC experiments, thermal measurements, and degradation analysis.
- Investigate electrical, thermal, and mechanical degradation effects to assess module performance, safety, and suitability for reuse in energy-storage applications.
- Analyze experimental data to support physics-informed modeling, state estimation, diagnostics, and remaining-useful-life assessment.

Publications

Journal Articles

- **F. Barat** and H. Fang, "System Identification of Nonlinear State-Space Models Using Ensemble Kalman Inversion with Application to Lithium-Ion Batteries," under review at *IEEE Transactions on Control Systems Technology*. [arXiv]
- M. Irani Rahaghi and **F. Barat**, "Solving Nonlinear Optimal Path Tracking Problem Using a New Closed-Loop Direct–Indirect Optimization Method: Application on Mechanical Manipulators," *Robotica*, vol. 37, no. 1, pp. 39–61, 2019. [DOI]

Conference Papers

- **F. Barat**, R. Carter, and H. Fang, "Robust State-of-Charge Estimation for Lithium-Ion Batteries by Neural Network-Enhanced Observer Design," submitted to *American Control Conference (ACC)*, 2027.
- **F. Barat**, R. Carter, S. Wilson, and H. Fang, "Recursive Parameter Identification of Nonlinear Stochastic State-Space Models via Sequentialized Ensemble Kalman Inversion," accepted to *IEEE Conference on Decision and Control (CDC)*, 2026. [arXiv]
- **F. Barat**, S. Wilson, H. Kim, and H. Fang, "System Identification of Lithium-Ion Battery Equivalent Circuit Models Using Ensemble Kalman Inversion," *American Control Conference (ACC)*, 2026. [IEEE Xplore] [arXiv]
- **F. Barat** and M. Irani Rahaghi, "Solving Nonlinear Optimization Problem Using a New Closed Loop Method for Optimal Tracking of a Path," *27th Iranian Conference on Electrical Engineering (ICEE)*, 2019. [DOI]

Teaching Experience

University of Kansas **Lawrence, KS**
Graduate Teaching Assistant *Spring 2025*

- Supported Numerical Analysis and Life Cycle Assessment courses; assisted students with MATLAB, numerical methods, engineering computation, and laboratory assignments.

University of Kashan **Kashan, Iran**
Graduate Teaching Assistant / Instructor *2015 – 2018*

Industry Experience

Diana Tajhiz Spadana Company / SEMAG Company **Isfahan, Iran**
Head of R&D / Mechanical Designer *Apr. 2018 – Jun. 2023*

- Led research-and-development and mechanical-design activities for industrial product-development projects from concept through detailed design.
- Designed mechanical components, assemblies, and engineering systems using **SolidWorks, Autodesk Inventor, AutoCAD, and CATIA**.
- Applied engineering analysis, numerical modeling, CAD, and design optimization to improve product functionality, manufacturability, and reliability.
- Coordinated technical development activities and collaborated with engineering, manufacturing, and project teams to resolve design and production challenges.
- Contributed to prototyping, design iteration, technical documentation, and implementation of engineering solutions for industrial applications.

Honors & Awards

- **3rd Place**, University of Kansas Graduate Student Research Showcase, 2025
- Ranked **4th** in M.Sc. Mechanical Engineering cohort, University of Kashan
- Ranked **1st** in B.Sc. Heat and Fluid Mechanics concentration, University of Kashan